

Q.1 (a) $y'' - 2y' - 8y = 0$

$$D^2 - 2D - 8 = 0$$
$$\Rightarrow D = \frac{2 \pm \sqrt{4 + 32}}{2} = \frac{2 \pm 6}{2} = 4, -2.$$

$\Rightarrow y(x) = C_1 e^{4x} + C_2 e^{-2x}.$

(b) $D^3 - 2D^2 - 4D + 8 = 0$

$$\Rightarrow D = 2, 2, -2$$
$$\Rightarrow y(x) = (C_1 + C_2 x) e^{2x} + C_3 e^{-2x}.$$

(c) $D^4 - 5D^3 + 6D^2 + 4D - 8 = 0$

$$\Rightarrow D = 2, 2, -1, 2$$
$$\Rightarrow y(x) = (C_1 + C_2 x + C_3 x^2) e^{2x} + C_4 e^{-x}.$$

2. $D^3 y - 3D^2 y - 4Dy + 12y = 0 \quad \text{--- (i)}$

Put $y = e^{mx}$ in (i), we get

$$m^3 - 3m^2 - 4m + 12 = 0$$
$$\Rightarrow m = 2, -2, 3$$

4. $\left| \frac{dy}{dx} \right| + |y| + 1 = 0$

$\underbrace{\left| \frac{dy}{dx} \right|}_{\geq 0} + \underbrace{|y|}_{\geq 0} + \underbrace{1}_{> 0} = 0$

$\Rightarrow |y'| + |y| + 1 \geq 1$

$\Rightarrow$ L.H.S. can't be zero.

5. $\left(\frac{dy}{dx} \right)^2 = 4y$

$\Rightarrow \frac{dy}{dx} = \pm 2\sqrt{y}$

Integrating $\Rightarrow 2\sqrt{y} = \pm 2x + C$

$\Rightarrow y = (x + C)^2 \quad \text{[general soln]}$

6. Method of undetermined Co-efficient

(a) $y'' - 2y' - 3y = 2e^{4x} \quad \text{--- (i)}$

Step 1 $\rightarrow$ solve $m^2 - 2m - 3 = 0 \Rightarrow m = 3, -1.$

$\Rightarrow$ C.F. = $C_1 e^{3x} + C_2 e^{-x}.$

Step 2:- Find particular :-

$$y_p = A e^{4x}$$

Put this in (i), we get

$$(16A - 8A - 3A) e^{4x} = 2e^{4x}$$

$\Rightarrow A = \frac{2}{5}$

$\Rightarrow y_p = \frac{2}{5} e^{4x}.$

$\Rightarrow y(x) = \text{C.F.} + y_p.$

(b) $y'' - 3y' + 2y = x^2 e^x$

$$m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$$

$\Rightarrow$ C.F. = $C_1 e^x + C_2 e^{2x}.$

Particular sol

Let

$$y_p = A x^2 e^x + B x e^x + C e^x.$$

$\Rightarrow y_p' = (2Ax + B) e^x + (Ax^2 + Bx + C) e^x$

$$= [Ax^2 + (2A + B)x + (B + C)] e^x$$

$y_p'' = [Ax^2 + (3A + B)x + (3A + 2B + C)] e^x$

Substitute in eqⁿ $\rightarrow$

$$[Ax^2 + (3A + B)x + (3A + 2B + C)] e^x - 3[Ax^2 + (2A + B)x + (B + C)] e^x + 2[Ax^2 + Bx + C] e^x = x^2 e^x$$

$\Rightarrow$ (i) $A - 3A + 2A = 1 \Rightarrow A = 1 \quad \text{[compare } Ax^2 \text{ coefficient]}$

(ii) $(3A + B) - 3(2A + B) + 2B = 0$

$\Rightarrow B = 3 \quad \text{[Comparing co-eff. of } x]$

(iii) $3A + 2B + C - 3(B + C) + 2C = 0$

$\Rightarrow C = 0 \quad \text{[By comparing constant]}$

Thus P.o.S. is $\rightarrow$

$$y_p = x^2 e^x + 3x e^x.$$

So,

Final solⁿ $y(x) = \text{C.F.} + \text{P.o.S.}$